

them using *Filter profiles*. Click *Filter profiles* under *HTTP proxy*.

11. Click *Add new*.
12. Enter the name of the profile. In our case, we used *Spam*. Click *Add and save changes*.
13. Click the button under *Configuration*.
14. To block all spam sites, let's use the *Threshold* option. The various options of *Threshold* will decide how to block the enlisted sites. Let's select *Very strict* under *Threshold* and click *Change*. Then click *Save changes* to save the changes permanently.
15. Select *Use antivirus* to block all incoming files, which may be viruses. Click the *Change* and the *Save changes* buttons.
16. To add a site to be blocked by proxy, click *Domain and URLs* and under *Domain* and *URL rules*, click the *Add new* button.
17. You will then be asked for the domain name. Enter the domain name of the site which is to be blocked. *Decision option* will instruct the proxy to allow or deny the specified site. Then click *Add* and *Save changes*.
18. To activate the *Spam* profile, click *Access rules* under *HTTP proxy*.
19. Click *Add new*. Define the time period and the days when the profile is to be applied.
20. Select *Any* from *Source* dropdown menu and then select *Apply filter profile* from *Decision* dropdown menu. You will see a *Spam* profile.
21. Click *Add* and *Save changes*.

With all the above steps, you will be able to either block or allow sites, depending on what you want your clients to have access to. All the other settings can be experimented with, as per your requirements.

Bandwidth throttling

The setting under *HTTP proxy* is used to add delay pools, so that a big file that users wish to download does not hamper the download speed of the other users.

To do this, follow the steps mentioned below:

1. First create the network object on which you wish to apply the rule. Click *Network* and select *Objects* under *Network options*.

2. Click *Add new* to add the network object.
3. Enter the name of the object, like *LAN*. Click *Add*, and then *Save changes*.
4. After you have added the network object, you have to configure members under that object. Click the icon under *Members*.
5. Click *Add new* to add members.
6. Enter the names of the members. We will use *LAN users*.
7. Under *IP address*, select the IP address range.
8. Enter the range of your DHCP address range, since we would like to apply it to all the users in the network.
9. Click *Add* and then *Save changes*.
10. Till now, we have added all the users of the network, on which we wish to apply the bandwidth throttling rule. Now we will apply the rule. To do this, click *HTTP Proxy* and select *Bandwidth throttling*.
11. This setting will be used to set the total amount of bandwidth that a single client can use. Click *Enable per client limit*.
12. Enter the *Maximum unlimited size per client*, to be set as a limit for a user under the network object. Enter '50 MB'. A client can now download a 50 MB file with maximum speed, but if the client tries to download a file of a greater size than the specified limit, the throttling rule will limit the speed to the maximum download rate per client. This speed option is set in the next step.
13. Enter the maximum download rate per client (for our example, enter 20). This means that if the download reaches the threshold, the speed will be decreased to 20 KBps.
14. Click *Add* and *Save changes*.

Traffic shaping set-up

With bandwidth throttling, we have set the upper limit for downloads, but to effectively manage our bandwidth we have to use the *Traffic shaping* module. Follow the steps shown below:

1. Click on *Traffic shaping* under the *Gateway* section.
2. Click on *Rules*. This will display two sections: *rules for internal interfaces and rules for external interfaces*.
3. Follow the example rules given in Table 1-- these can be used to shape the bandwidth on *eth1*.

Table 1

Based on the firewall	Service	Source	Destination	Priority	Guaranteed rate (KBps)	Limited rate (KBps)
Yes	Application-based service group -> Service Audio	Any	Any	2	512	0
Yes	Application-based service group -> Service Video	Any	Any	1	512	0
Yes	Port-based service -> HTTP	Any	Any	3	1024	2048
Yes	Port-based service -> HTTPS	Any	Any	3	1024	2048
Yes	Application-based service group -> p2p	Any	Any	7	0	10